

'SILICON VALLEY' LOCATIONAL CLUSTERS: WHEN DO INCREASING RETURNS IMPLY MONOPOLY?

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It is often argued that the presence of increasing returns implies that one product, or one technology, out of several possible must come to dominate a market. This paper examines this argument in the context of regional economics. It constructs a model of industry location where firms choosing among regions are attracted by agglomeration economies due to the presence of other firms in these regions, and where 'historical accident' enters because firms are heterogeneous and enter the industry in random order. It asks: When do economies of agglomeration lead to a 'Silicon Valley' – a single dominant location or region monopolizing the industry?

The paper shows that: (i) Where there is no upper bound to locational increasing returns due to agglomeration, there will indeed be a monopoly outcome: industry will cluster in one dominant region, with probability one. (*Which* region depends both on geographical attractiveness and accidental historical order of firm entry.) (ii) Where there is an upper bound to increasing returns due to agglomeration, certain sequences of firm entry can produce dominance by one region; others can produce regional sharing of the industry exactly as if the agglomeration effects were absent.

Construed more generally in economics, the results show that increasing returns, if bounded, do not guarantee monopoly outcomes.

Key words: Path-dependent dynamics; location theory; stochastic processes; strong laws; increasing returns; non-convex economics; monopoly; regional disparity; economic geography.

1. Introduction

It is often argued in economics that the presence of increasing returns – decreasing costs, or network externalities – implies that one product or one technology out of several possible must come to dominate a market. This paper examines this argument in the context of industrial location and regional economics.

Increasing returns occur in regional economics in the form of agglomeration economies – net benefits of firms being close to other firms. Where firms entering an industry benefit from the presence of other firms, a predominant share of the industry may cluster in a single region, or district, or location – a 'Silicon Valley'

– not necessarily because of any intrinsic advantage of that particular location, but because 'historical accident' placed certain firms there initially and this concentration of firms in turn attracted a high proportion of subsequent entrants.

This argument, that a single region may come to monopolize an industry due to agglomeration economies combined with 'historical accidents', is not new. It goes back to the early decades of this century. Several of the less-known members of the German School of regional economics, in particular Engländer (1926), Ritschl (1927), and Palander (1935), viewed observed locational patterns not as the solution to a unique-equilibrium spatial-economic problem, but rather as the result of an historically-dependent process; with new industry attracted by existing agglomerations, laid down stratum by stratum upon inherited, previous locational formations, and often clustering where early firms put down.¹ Their viewpoint never quite took hold, however, because it was never properly articulated. Agglomeration economies introduced analytical difficulties: when one or a few regions might end up with the entire industry, *which* regions might come to dominate, as Isard (1956) noted, was not determinate. The problem was non-convex and there were multiple equilibria. How one equilibrium – one regional pattern – came to be 'selected' was not at all clear when there were multiple candidate regional patterns of industry location.

This paper attempts to resolve the problem of industry location under agglomeration economies by asking two main questions: (i) Under what circumstances do economies of agglomeration lead to the emergence of a 'Silicon Valley' – a single dominant location or region? (ii) Where agglomeration economies cause there to be multiple asymptotic 'solutions' to the locational problem, how does one solution come to be 'selected'? It sets up a model of the dynamics of industrial location that allows 'historical accidents' in the shape of heterogeneous firms entering the industry in random order, and where regions with higher proportions of the industry exert agglomeration attraction on subsequent entrants.

The results show that increasing returns to agglomeration, if unbounded, indeed cause monopoly by a single region or location, and that 'historical accident' is all-important in selecting *which* region achieves dominance. However, increasing returns, if *bounded*, do not necessarily cause a 'Silicon Valley' monopoly outcome: where there are upper limits to the agglomeration benefits, certain combinations of 'historical accident' can produce a single dominant region; others can produce regional dispersion of the industry exactly as if agglomeration effects were absent.

¹ By this argument, Santa Clara County in California, the actual Silicon Valley, has a high concentration of the U.S. electronics industry not necessarily because it is close to Pacific sources of supplies or has better access to advances in academic engineering research than elsewhere, but rather because certain key people – the Hewletts, the Varians, the Shockleys of the industry – happened to set up near Stanford University in the 1940s and 1950s, and the local labor expertise and inter-firm market they helped to create made subsequent location in Santa Clara County extremely advantageous for the thousand or so firms that followed them (see Cohen, 1984). Similar arguments appear in the regional-disparity literature: Myrdal (1957), Faini (1984); and in the 'growth pole' theories of Perroux (1955) and others in the 1950s. Allen and Sanglier (1981) use simulation techniques to show the importance of agglomeration economies in determining locational patterns.

And, paradoxically, agglomeration effects can easily *cause* regional separation of the industry.

2. A model

First, some preliminaries. I will say there are *economies of agglomeration* (Weber, 1909; Hoover, 1937) if net benefits to being in a location together with other firms increase with the number of firms in the location. The sources are several: as a location gains firms, it also gains useful infrastructure. Its labor markets deepen (David, 1984a). Specialized legal and financial services appear. Spare parts and out-of-stock items become locally available, reducing inventory costs. And social networks come into being where information, expertise, and contracts can be easily exchanged. *Diseconomies of agglomeration* – net benefits that decrease with increasing numbers of other firms – may also occur. The location may become congested; land may become expensive; or firms that sell locally may find themselves in monopolistic competition with nearby firms as in Lösch (1941). And of course, besides agglomeration effects, there may be benefits or costs due to geographical differences – in transportation cost, in the availability of raw materials, in factor prices, in climate, or in proximity to markets.

I now set up a model where firms in an industry decide on locating in one of N possible regions, or sites, or cities. Each firm is well informed on profits or returns – the net present value of siting in each location – at the time of its choice; each chooses what, for it, is the maximum return location and then stays there. I assume that new firms are added slowly and the future is discounted so that initially we can ignore expectational problems and assume that new firms decide to locate at their sites of *present* maximum return.

The firms in an industry differ in product mix or in manufacturing process – hence in locational needs. In this simple model I condense all benefits or returns, r_j^i , to firm i for locating in site j , into two components:

$$r_j^i = q_j^i + g(y_j), \quad (1)$$

where q_j^i is the *geographical benefit* to firm i for siting in location j (returns in the absence of other firms); and $g(y_j)$ is the net *agglomeration benefit* from having y_j firms already located there at the time of choice. (I will speak of firm *tastes*, understanding these to be geographical preferences in the absence of other firms' settlement. Where the agglomeration function, g , is increasing, we have economies of agglomeration; where decreasing, diseconomies of agglomeration.)

Firms enter the industry at different times. Thus, we may think of them as arriving in sequence to make their choice. I introduce 'historical accident' by supposing that the sequence of firm types that enter is unknown to an observer. Thus, we can represent each new firm as a random 'drawing', of a vector $q = (q_1, q_2, \dots, q_N)$ which denotes that firm's locational tastes for each possible site, from a given

distribution F of potential firms over locational tastes. (Thus, F determines the frequency of arrival of different firm types, and in general there is a continuum of possible firm types.) I assume that the vector of tastes, q , has finite support. Note that since we can add a positive constant to all q 's without changing location decisions, we can treat q as always positive. Sites that are frequently preferred (in the absence of agglomeration economies) I will call *attractive* and I include in the analysis only sites that would be chosen with positive probability.

This model has been kept deliberately simple so as not to complicate the subsequent analysis. It could be modified to specify how new firms are generated (see Arthur, 1988) and how other industries might affect locational returns. A later section discusses several extensions.

Given the above assumptions, we can compute the probability that an entering firm chooses location j . Suppose there are $y_1, y_2, \dots, y_N$ firms already in locations 1 through N , respectively. Then the probability that the next firm 'drawn' prefers j over all other sites is:

$$p_j = \text{Prob}\{[q_j + g(y_j)] > [q_i + g(y_i)] \text{ all } i \neq j\}. \quad (2)$$

We can calculate this probability as the (Lebesgue-Stieltjes) integral:

$$p_j = \int_{V_1} \dots \int_{V_j} \dots \int_{V_N} dF(z_1, \dots, z_j, \dots, z_N), \quad (3)$$

where the region of integration $V_i = (-\infty, z_j + g(y_j) - g(y_i)]$ for $i \neq j$ and $V_j = (-\infty, \infty)$. I will call $p = (p_1, p_2, \dots, p_N)$ the probability vector of locational choice. Notice that through the agglomeration function g these probabilities p of an addition to locations 1 through N are a function of the current configuration of the industry, as we would expect.

The problem is now to derive and examine the locational pattern – the proportion of the industry in regions 1 through N – under different assumptions about geographical benefits and increasing returns due to agglomeration economies.

3. Dynamics of the locational process

We proceed by tracing the locational pattern of the industry as it forms, and as firms of different types enter it. We would expect this pattern to fluctuate somewhat because of the randomness in entry of firm type; and we would expect its movement to depend upon its present state, because locational choices depend on the current placement of firms.

Starting from a given number of firms in each location (which, for convenience, we set at zero), the industry forms by the addition of one firm at a time, distributed to one and only one location. We can describe the spatial configuration of the industry at 'time' n , after n firms in total have located, by the vector of firms in the N locations $Y_n = (y_1(n), \dots, y_N(n))$. An equivalent but more convenient description

is the vector of *locations' shares* or proportions of the industry, $X_n = (x_1(n), \dots, x_N(n))$, where $X_n = Y_n/n$. The vector of probabilities of locational choice, p , is a function of Y or, equivalently, a (continuous) function of the locational shares of the industry, X , and of n .

Now, the industry forms by the addition of one firm at a time, so that the vector of firm numbers in the potential locations evolves as:

$$Y_{n+1} = Y_n + b(n; X_n); \quad Y(0) = 0, \quad (4)$$

where b is the j th unit vector with probability $p_j(n; X_n)$.

Then, dividing (4) by $n+1$, the vector of proportions of the industry in sites 1 through N , or locational shares, evolves as:

$$X_{n+1} = X_n + \frac{1}{n+1} [b(n; X_n) - X_n]; \quad X_0 = 0. \quad (5)$$

We can rewrite (5) in the form:

$$X_{n+1} = X_n + \frac{1}{n+1} [p(n; X_n) - X_n] + \frac{1}{n+1} \mu(n; X_n), \quad X_0 = 0, \quad (6)$$

where μ is defined as the random vector:

$$\mu(n; X_n) = b(n; X_n) - p(n; X_n).$$

Equation (6) is the description of the dynamics of locational shares that we want. Mathematically it has the form of a stochastic approximation.

Notice that the conditional expectation of μ with respect to the current state X_n is zero, hence we can derive the *expected motion* of the locational shares as:

$$E(X_{n+1} | X_n) - X_n = \frac{1}{n+1} [p(n; X_n) - X_n]. \quad (7)$$

We see that, if the probability $p_j(X_n)$ of an addition to location j is greater than the current proportion of the industry $x_j(n)$ in location j , this location's share should increase – at least on an expected basis. Conversely, if the probability is less than the current proportion, it should decrease.

Equation (6) therefore tells us that locational shares are driven by an expected motion effect (the second term on the right) that derives from locations' relative attractiveness and current agglomeration pull, together with a perturbation effect (the third term) that derives from randomness in entry. Both expected motions and fluctuations have less effect on shares as the industry grows. Both die away at the rate $1/n$; hence we might expect locational shares to settle down as the industry becomes large.

We can get some idea of how locational shares might settle down, for a given locational probability function p , if we plot expected motions on the unit simplex of locations' shares (as in Fig. 1).

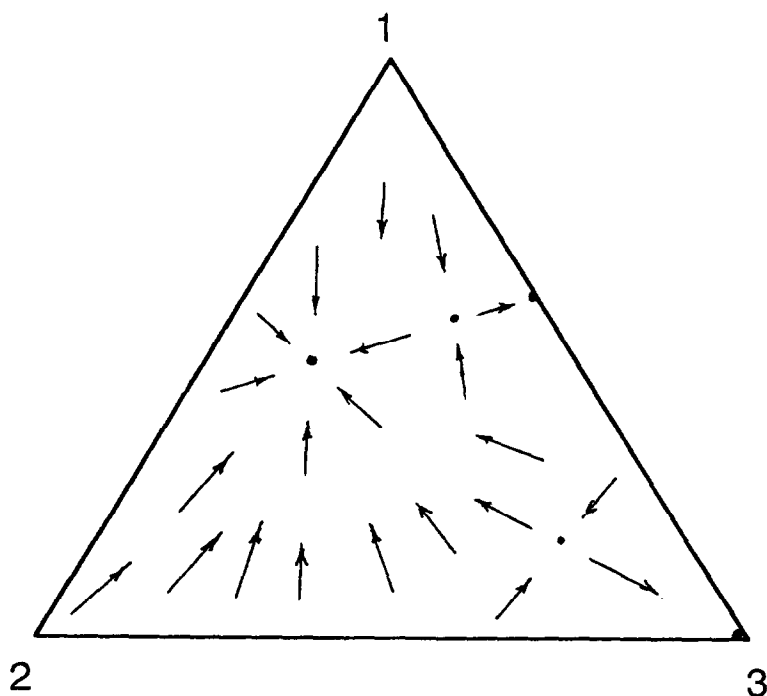


Fig. 1. Expected motions for a given locational probability function.

Depending upon the function p – and ultimately on the distribution of firm tastes and the agglomeration function – there may be several points x (several locational patterns) at which expected motion is zero. These are the fixed points of p , where $p(x) = x$. Some of these are attractor points, some are repellor points. We might conjecture that, if choices in a given problem are ‘driven’ by a locational probability-of-choice function p , the locational pattern will converge to a fixed point of p on the simplex of locational shares, and to one that is an attractor. Furthermore, if there are multiple attractors, they might all be candidates for the limiting locational pattern, with the locational distribution of the industry falling into the orbit of one of these by ‘historical accidents’ of sequence of entry.

A recent set of theorems for path-dependent processes like (6), due to Arthur, Ermoliev and Kaniovski (1983, 1986, 1987a, 1987b) (referred to here as AEK and applicable to a broad variety of nonconvex problems in economics and physics) confirms these conjectures. Under certain technical conditions:² (i) X_n converges with probability one to a fixed point where $p(x) = x$; (ii) X_n can converge to ‘attractor’ or stable fixed points (points on the simplex that expected motions are toward);

² AEK require that the set of fixed points of p have a finite number of connected components. Recent work (Arthur, Ermoliev and Kaniovski, 1987b) allows non-unit increments and discontinuities in p . For a non-technical introduction with applications in various fields see Arthur, Ermoliev and Kaniovski (1987a). The $N=2$ case was analyzed by Hill, Lane and Sudderth (1980).

and (iii) X_n cannot converge with positive probability to 'repellor' or unstable fixed points (points on the simplex that expected motions lead away from). Where the function p varies with time n as in our locational case, this generalized strong-law theorem still goes through, provided that the dynamics 'settle down' to stationarity – that is, provided that the set of 'driving' probabilities $\{p_n\}$ tends to a limit function p .

Notice that the standard strong law of large numbers makes statements about limiting shares when increments are added independently of present state of the process. AEK is thus a generalization of the strong law of large numbers for the case of path-dependent increments.

4. Locational patterns in several cases

I have described the spatial formation of the industry as a process where unit increments (firms) are added to one of N categories (candidate locations) with a probability vector p (of an addition to category 1 through N) that in general is a function of the current proportions X_n in the N categories: $p = p(x; n)$. We now seek limiting vectors of locational shares, and we start with a trivial but useful benchmark case.

Suppose agglomeration economies are absent, so that g is zero and only geographical considerations q count. We see from (3) that the probabilities of choice p , are constant and are therefore independent of locational shares at all times. Thus, increments are added independently of current proportions, and we can invoke the standard strong law of large numbers to show that the locational shares X_n converge to p . We then have

Theorem 1. *Where there are no economies or diseconomies of agglomeration (that is, $g \equiv 0$), then, with probability one, locational shares tend to fixed, predictable proportions in the limit as the industry becomes large.*

In this 'pure attractiveness' case, locational dispersion of the industry takes place to the degree the industry is heterogeneous in taste. If all firms are alike, each will choose the same maximal location and the industry will cluster. Thus, dense concentrations of the film industry near Los Angeles may not necessarily be evidence of the presence of agglomeration economies. They may simply reflect homogeneity of early needs – for cheap real estate, dry weather, and good outdoor lighting. Agglomeration economies are not a necessary condition for spatial clustering.

4.1. Unbounded agglomeration economies

In our more general case where agglomeration economies *are* present, increments to locations are *not* independent – the probability of adding firms depends upon past additions – so that the standard strong law is not usable. We now examine this

case assuming first that additions of firms always confer net benefits on a location, without upper limit. We might expect that if one location gets sufficiently ahead in industry share by 'historical accident', it will build up an unassailable advantage and eventually monopolize the industry. This indeed happens.

Theorem 2. *Where there are unbounded economies of agglomeration such that the agglomeration function increases at least at the rate $\partial g/\partial y > \varepsilon y^\mu$, where $\mu > -1$, then, with probability one, in the limit as the industry becomes large, one location takes all but a finite set of firms. Locational shares tend to zero in all but one location.*

Proof. The argument consists in showing that if n is large enough, for points in the simplex of locational shares the location of highest share must be chosen with probability one. Thereafter that location is repeatedly chosen; the process is locked-in to it and its share of the industry goes to 100%.

Consider the set $B = \{x \in S^N: \text{two or more elements of } x \text{ are maximal}\}$ (see Fig. 2); and B_ε , its ε -neighborhood. Points in $S^N \setminus B_\varepsilon$ then represent locational shares with a unique maximal element, with index J say (of the nearest vertex). Partition $S^N \setminus B_\varepsilon$ into separate sets C^J designated by the nearest vertex J . Now consider a given point z in C^J . The fact that g is increasing and unbounded guarantees that there exists a finite time $n(z)$ such that returns $(q_J + g(n(z)) \cdot z_J) > (q_I + g(n(z)) \cdot z_I)$,

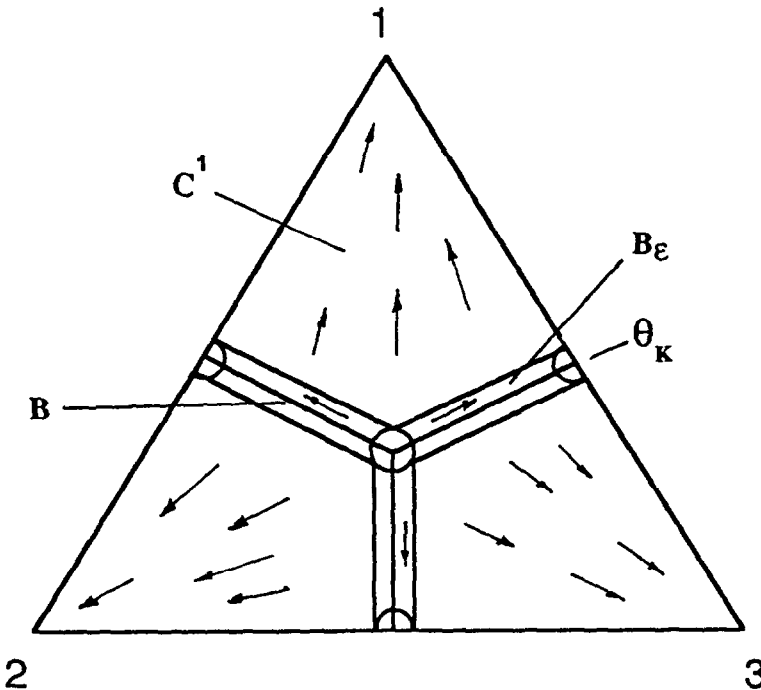


Fig. 2. Expected motions under unbounded agglomeration economies.

all $I \neq J$. Furthermore, it is easy to show that $n(z)$ has an upper bound n^J in C^J . Therefore, if the process X_n is in C^J at a time n greater than n^J , location J will have maximal returns for all firm types; it will then be chosen with probability one; it will continue to be chosen with probability one; the process will never exit C^J and it will converge to 100% share for location J .

Alternatively, the process might not enter a set C^J ; it could stay within B_ϵ forever. Consider 'balanced' points θ_K defined to be points in the simplex where locations outside the subset K have zero proportion and those inside K have equal proportions. No 'unbalanced points' in B_ϵ can be sustained: beyond some finite time locations with least share will be shut out as before. Furthermore, the process cannot cycle: unbalanced points cannot be recovered once they are shut out. X_n can then stay within B_ϵ indefinitely only by converging to a 'balanced point' θ_K . However, direct calculation shows that, under the condition stated for g in the theorem, balanced points θ_K are *unstable*. That is to say, we can show that as long as $\partial g / \partial y > \epsilon y^\mu$, with $\mu > -1$, then any given θ_K satisfies the instability criterion that there exists within K a symmetric matrix C , and parameters λ and $\nu > 0$, such that $\langle C_{p_n}(z), z - \theta_K \rangle \geq \lambda \langle C(z - \theta_K), z - \theta_K \rangle^{1+\nu} + \delta_n$, where $\sum_n \delta_n < \infty$, for z in a neighborhood of θ_K . By Theorem 3 of Arthur, Ermoliev and Kaniovski (1989), the instability of θ_K ensures that $\text{Prob}\{X_n \rightarrow \theta_K\} = 0$. The process cannot converge to θ_K . It must exit B_ϵ in finite time and must therefore converge to some vertex J . The theorem follows.³ Notice that the limiting probability function p becomes a step function (with element $p_J = 1$ for points with nearest vertex J and with each vertex an attracting fixed point). $\square$

In this unbounded-agglomeration case the locational process eventually 'locks in' to a 'Silicon Valley' outcome – a monopoly of one location. *Which* location this is, of course, depends on the order of entry of firm types and very much on early locators' relative preferences for locations. The process is path-dependent (more precisely, non-ergodic) in that a slightly different order-of-choice history early on could sway the outcome to a different location becoming predominant. An attractive location will likely be favored by many firms early in the choice order, and therefore it has a larger probability of predominating. Attractiveness, interacting with historical accidents of choice-order, determines the outcome.

It may seem obvious that monopoly must result in this unbounded agglomeration case. Yet it is easy to construct entry-sequences that lead to locations sharing the industry indefinitely; and to show that there is an infinity of such sequences. The power of this theorem lies in its showing that such nonmonopoly outcomes have measure zero in the sample space of all possible outcomes.

³ In reality, of course, industries do not grow forever. It is best to construe this theorem as a law-of-large-numbers result in the usual sense that the larger the eventual size of the industry, the more likely locational shares will lie in the neighborhood of a vertex.

Remark 1. Where the density function f of geographical preferences has infinite 'tails' in all directions instead of bounded support, it is easy to check that the vertices remain the only stable attractors, and the above theorem again goes through: a single location again monopolizes.

4.2. Bounded agglomeration economies

In our next case, where the increasing returns – the agglomeration economies – are *bounded*, monopoly no longer occurs with probability one. In this case, locations could, with positive probability, share the industry exactly as if agglomeration economies were absent.

To see why this should be so, suppose agglomeration economies g are increasing but bounded above by the constant h : thus, adding firms enhances a location's advantages, but not without limit. Suppose two regions only. Now it may happen that if enough firms that prefer region A occur early in the queue of entries, A will move sufficiently ahead to dominate B for all subsequent choosers. As before, one location will take all. Suppose, however, that firms preferring A and firms preferring B arrive more or less evenly in the sequence. Then regions A and B will be bid upward in tandem; both regions will reach the agglomeration upper bound and neither will be sufficiently ahead to shut out the other (see Fig. 3). But now the agglomeration economies realized will be equal; they will cease to matter in locational choice and so eventually both regions will share the industry, with shares that will equal the proportion of firm types that prefer A and B in the absence of agglomeration economies. Whether monopoly or locational sharing takes place depends, of course, on 'historical accident' of the entry sequence.

Our N location case is more complicated. We must first determine the 'candidate' sets of locations that if bid upward together could shut the other locations out and thus together dominate the industry.

Consider the (random) vector of returns $\pi = q + g(y)$, for a given vector y of firms. Important in what follows is the *support* of these returns, which we will designate Q_y . (The support Q_y indicates the region in N -dimensional preference (or returns) space over which firm types are spread.) As settlement takes place and agglomeration economies accrue to locations that are chosen, the returns support Q_y will shift. If it enters a region where some subset of locations always have maximum returns, all firms will choose locations in this subset, other locations will never be chosen, and the process will be then locked-in to this subset of locations.

Now, let K be a set of locations, and let M_K be the cone of vectors in $\mathcal{R}^N$ with maximal element in the set K . Let Q^K be the support of the returns vector $q + \gamma_K$, where the vector γ_K has 0-elements for locations not in K and h -elements for locations in K . (This is where returns would eventually lie if locations in set K only were to receive firms.) We will say that K is a potential *lock-in set* if (i) $Q^K \subset M_K$ and (ii) $Q^K \not\subset M_{K'}$ for any $K' \subset K$. We will say that K is a *potentially dominant set* if there also exists some finite sequence of arrivals that causes Q_y to belong to $Q_\varepsilon^K \subset M_K$, where Q_ε^K is any given ε -neighborhood of Q^K .

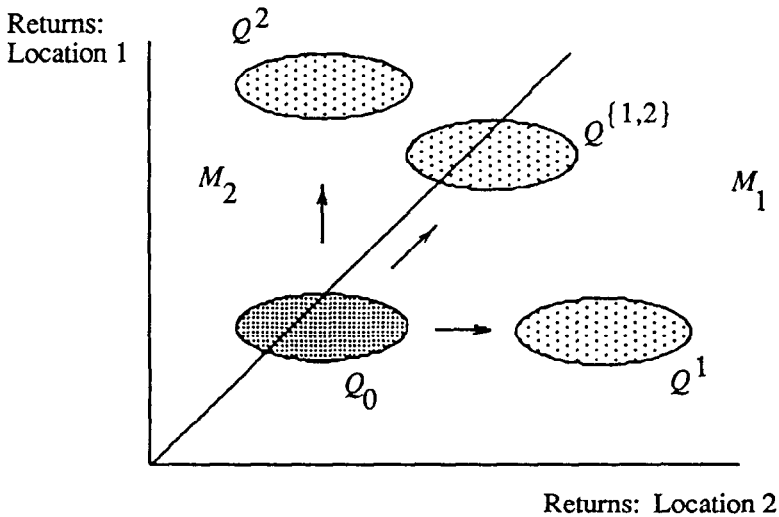


Fig. 3. Support sets of firms' returns for a two-location problem.

Theorem 3. *Suppose economies of agglomeration g are increasing and bounded above. Then in the limit as the industry becomes large, one set of locations from the collection of potentially dominant sets will take 100% of the industry, with probability one, and each such set K has positive probability of being this monopolizing set. Furthermore, locations in the monopolizing set share the industry exactly as if no agglomeration economies had been present and they were the only locations available at the outset.*

Proof. First, the process must converge with probability one to non-zero proportions in some subset J . One firm is added each time, and so some set J of locations must continue to receive firms ad infinitum. Locations in this set would be bid to their agglomeration ceiling. Probabilities of additions to locations in J therefore become independent of current proportions; and by the strong law of large numbers, the process must converge.

Now suppose the process converges to non-zero proportions in a non-potentially-dominant set J . The non-dominance of J implies either that some locations in J receive firms when their locational probability is zero, or that choices outside J occur with positive probability. Either way convergence to J is contradicted.

Finally, consider a set K of locations that is *potentially dominant*. By definition, there may occur with positive probability a finite sequence of arrivals that causes returns Q_y to lie within Q_e^K in the cone M_K . Once returns lie within this set, locations other than those in K cannot be preferred and the process locks in to this set K . Therefore the set K can monopolize the locational process with positive probability, with location proportions X_n converging to the relative proportions of firms that prefer each location within K . The theorem follows. $\square$

In this bounded agglomeration case, industry can be shared by several locations. The reason is not so much that agglomeration is overpowered by heterogeneity, but rather that certain choice-orders cause several locations to accumulate firms at more or less the same rate, and therefore agglomeration forces remain balanced enough as the industry forms for heterogeneity of tastes to spread the firms over these locations. Thus, the monopolizing tendencies of increasing-return effects may be overcome by even a small degree of heterogeneity. Historical choice-order once again matters greatly here – it 'selects' the set of dominant locations that will share the industry. But once this set is decided and 'locked-in', the locational process proceeds as if the agglomeration effects were entirely absent. There are two extreme cases worth mentioning. First, agglomeration forces may be too weak to cause any monopolizing of the industry by subsets of locations at all:

Corollary 1. *If $h < \text{Min}_K(d(Q_0, M_K))$ then all locations share the industry, with probability one, in the proportions that would have occurred if agglomeration economies were absent.*

(Here d is the set-distance operator.) In this case potential agglomeration economies are not great enough to overcome heterogeneity of preferences. Equivalently, there are always some firm types that do not switch their 'natural' choice to the populated locations. All locations eventually reach the agglomeration ceiling, and so agglomeration forces may as well not be present at all. In this special case history does not matter.

Second, conditions might exist under which a single location *must* monopolize the industry even though agglomeration effects are bounded. To illustrate this, let Q_e^j be the support of the returns when one firm only has chosen and is in location j .

Corollary 2. *If $Q_e^j \subset M_{\{j\}}$ for all j , then one location takes all the industry with probability one.*

In this extreme illustrative case agglomeration economies drive the support straight into the dominance region of the location that receives the first firm. Practically speaking, this can happen when preferences are tightly clustered, and when agglomeration effects at the outset are strong.

In the results above, it was crucial that firms' tastes had bounded support. Suppose now they have unbounded support – that is, that the distribution function F of geographical preferences has infinite tails in all directions.

Theorem 4. *Where preferences F have unbounded support, and the agglomeration function g is bounded above, then all locations share the industry, with probability one, in the proportions that would have occurred if agglomeration economies were absent.*

Proof. In this case no subset can ever become dominant – there is always a positive probability of a location outside the subset being chosen. Locations are added until all have been bid up to their agglomeration bound. We may subtract this agglomeration advantage equally from all locations; the dynamics then proceed as if no agglomeration component were present, and all locations share the industry according to their natural probabilities of being chosen. $\square$

5. Locational orphaning and spatial separation

Many industries are not concentrated, Silicon Valley fashion, into a dominant location. Firms are spread out with some spatial regularity. The usual, Löschian explanation for spreading and separation invokes spatial monopolistic competition: firms distance themselves from their rivals for local markets. That is, *diseconomies* of agglomeration cause firms to separate spatially.

Strangely enough, *economies* of agglomeration can also cause separation and spreading. The reason is that locations with large numbers of firms are preferred to geographically similar, neighboring locations with few firms. Locations with large numbers of firms therefore cast an 'agglomeration shadow' in which little or no settlement takes place. This causes separation of the industry.

To see this in more detail, consider now a two-dimensional landscape with returns q for any given firm-type varying continuously over the surface – reflecting climate variation or transportation costs, say. (To preserve finiteness of the number of locations we can treat potential locations as points on a fine grid. The industry is still heterogeneous, so that tastes q are distributed as before in the density function f .) Tastes are uniformly continuous in the sense that (for all firm types) tastes for two neighboring sites always differ by less than some ε .

Consider two locations, a and a' , separated by d units (under a suitable metric), having k and k' firms respectively. Location a , with more firms, will dominate location a' for *all* firm types if $g(k) + q(a) > g(k') + q(a') + \varepsilon \cdot d$, so that site a' will not receive any future firms. When attractiveness varies smoothly over the landscape, then places near a location with a larger number of firms are similar to this location geographically; and they are not sufficiently more attractive to overcome the agglomeration advantage of their dominant neighbor. These places have become dynamically *orphaned*, and we can say that they lie within the *agglomeration shadow* of their dominant neighbor.

As a location receives firms its agglomeration shadow grows; it shuts out further neighbors and it thereby enlarges the domain of tastes – the 'gravitational region' – from which it receives further firms. The result of this, in the realistic case where agglomeration economies are bounded, is that neighboring locations cannot share the industry but that sufficiently separated regions can.

Agglomeration economies have *caused* separation and dispersion. And once again *which* locations gain the industry and which become orphaned is a matter of historical accident.

6. Remarks and extensions

I have shown in a simple model that where agglomeration economies are present, historical 'accidents' of choice-order select a limiting spatial pattern from a possibly large collection of candidate spatial patterns. The following are brief remarks on various possible extensions:

(1) Suppose there were eventual net *diseconomies* of agglomeration, so that disamenities of industry ultimately outweighed agglomeration benefits. In this case g would eventually decrease and be unbounded below. Locations would fill up, partly by historical accident, the attractive ones first, until diminishing returns set in. Then industry would begin to site in the less attractive locations, and still less attractive, until all locations showed diminishing returns. I leave it to the reader to confirm that the point $x = (1/N, \dots, 1/N)$ would be the only attractor of the limiting locational probability function, and therefore that the industry would eventually be shared equally between all locations, with probability one. Under diseconomies of agglomeration, historical accident would not count.

(2) Suppose firms were affected not just by the presence of other firms only in the same locality but by the entire configuration of firms in *all* other locations. Total returns would then become:

$$r_j^i = q_j^i + g(y_1, y_2, \dots, y_N), \quad (8)$$

with the wider agglomeration function g also possibly location- or firm-type-specific if necessary. This form would allow cases of more general agglomeration or deglomeration effects (such as the Löschian monopolistic-competition case where neighboring competitors cut into a firm's demand to a degree that varies with distance). Analysis in this entire-configuration framework would be similar to that above – the AEK strong law for example handles this case directly – but calculation of the probabilities of locational choice would now be more complicated. I would expect findings similar to the above.

(3) Suppose transaction costs to moving were zero, so that firms could move, and thus 'recontract' their locational decisions. Now after each new entrant's choice there would be a round of relocations of settled firms that could improve their returns. Firms would tend to move, from places they settled in that received firms early on but which have now by 'chance' become orphaned or left behind in agglomeration, to places that are currently heavily populated with firms. Again there would be multiple potential outcomes, with 'accidents' of choice-order in moving as well as in entry selecting an ultimate locational pattern. 'Historical accident' would still count.

(4) Suppose entering firm-types were not independent, but that there was sequence 'linkage': certain entrants might attract complementary producers; later entrants might differentiate their firm-type to fit with earlier entrants; and particular stages of an industry's technological development might be associated with the entry

of a characteristic type of firm. Firm-type arrivals would then be serially correlated; and path-dependence would extend now to firm-types and to recent arrivals, as well as to previous locational choices. It would be simple to formulate this case, but analysis would require more general path-dependent techniques than the ones used here.

(5) Suppose present choosers were affected by future firms' location decisions. We might then assume that firms form beliefs about the type of stochastic process they find themselves in, in the shape of conditional probabilities of future states of the locational process. Expectations would therefore change as the settlement process worked itself out. In choosing, firms would create an *actual* stochastic process, which would be in rational expectations equilibrium if it were identical to the *believed* stochastic process (cf. Lucas, 1972). Preliminary results (see Arthur, 1989) show that expectations would exacerbate monopoly tendencies. An unattractive location that was lucky early on could come to dominate as choosing firms increasingly enhanced their beliefs of that location's success.

7. Conclusion

This study investigated conditions under which increasing returns in the form of agglomeration economies cause monopoly – in this case 'Silicon Valley' clustering. It showed that if agglomeration economies are unbounded, then a single region or location selected by 'historical accident' indeed monopolizes the industry. But contrary to conventional beliefs, agglomeration economies are not sufficient to guarantee a single-location dominant outcome. If agglomeration economies are bounded, even a small amount of heterogeneity can cause regions to share the industry as if agglomeration economies were absent. Increasing returns do not necessarily imply monopoly.

If locational increasing returns are present in the industry location problem, static equilibrium analysis is not enough. Settlement patterns are path-dependent; we are forced to follow the locational structure as it forms, and this requires dynamic analysis. Interestingly, the picture that emerges is more Darwinian than equilibrational in nature, with geographical attractiveness bestowing 'selectional advantage'; locations 'selected' early on by chance becoming locked-in or 'fixed'; and some locations exercising 'competitive exclusion' on others by appropriating the available agglomeration economies.

The locational analysis of this paper would apply to both the product differentiation and standardization literature, in the case where products like Sony Betamax® or standards like the ASCII code enjoy network externalities in the sense that their appeal widens as their market share of user-base grows (see Farrell and Saloner, 1985; Katz and Shapiro, 1985; and Arthur, 1989). We might ask under what circumstances heterogeneity of taste for differentiated products or for competing standards might overcome such network-externality increasing returns, to prevent

monopoly and therefore loss of product diversity, or to prevent emergence of a de facto, possibly inferior standard. Many of the results would transfer directly.⁴

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⁴ They transfer most directly to the case of competition between differentiated products that are not priced strategically, or between generic, 'non-sponsored', competing standards. Strategic maneuvering would correspond in our problem to locations offering incentives, in what David (1984b) calls a 'locational tournament'. This would add a game-theoretic overlay.

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